Requirement Engineering:

It is the process of identifying, eliciting, analyzing, specifying, validating and managign the needs. and expectations of stakeholders for a software system.

Step of Requirement Engineering:

1. Feasibility study
2. Requirements Elicitation
3. " specification
4. " verification and validation
5. " Management.

1. Feasibility:

It checks whether the project is feasible or not. It evaluates technical, operational, economic, legal and schedule aspects of a project.

(2) Requirement elicitation:

It focuses on gathering stakeholder needs and domain knowledge about the project. Interview, survay, observation, prototyping are some of the method to collect requirement.

(3) Requirement specification:

The process of documenting the requirements identified in the requirement elicitation step in a clear, consistent and unambiguous manner. This step also involves prioritizing and grouping the requirement into manageable chunks. Uses model like DFD, ER diagram etc.

* Functional Requirements: These describe what the software system do. They define the specific behaviors, tasks or functionals the system ~~as~~ must be able to perform.

Ex: In a e-commerce app, a function would be: The customer must add items to shopping cart and proceed to check out.

- **Non-functional requirement:**

These describe how well the software system should do it. They specify the quality attributes, performance, security or usability of the system.

Ex: In the same e-commerce application

- The shopping cart page must loaded in under 2 sec, even when 10000 users are accessing the site simultaneously.

- **constraint:**

These describe any limitations or restrictions that must be considered when developping the software system.

Ex: Use personal data must be encrypted

- **Acceptance criteria:**

These describe the conditions that must be met for the software system to be considered complete and ready for release.

Ex: The user must enter a valid registered email address and password.

Requirement verification and validation:

verification:

It is checking that the requirements are complete, consistent and accurate. It involves reviewing the requirements to ensure that they are clear, testable and free of errors and inconsistencies. The can include reviewing the requirements document; models; diagrams and holding meetings.

validation;

It is the process of checking that the requirements meet the needs and expectation of the stakeholders. This can include testing the software system through simulation, testing with prototypes and testing with the final version of the software.

Requirement management:

The process of analyzing, documenting, tracking, prioritizing on the requirement and controlling the communication with relevant stakeholders. This stage takes care of the changing nature of the requirements. Tracking and controlling changes, version control, communication, monitoring and reporting.